

CS101-Quiz5-Review

Suting Chen

CS101-Quiz5-Review

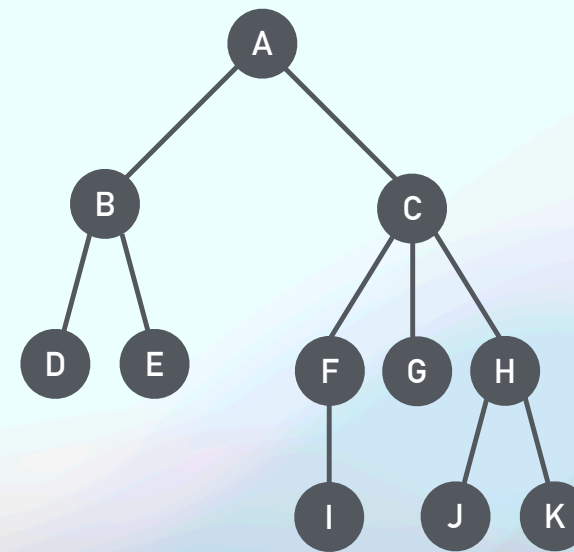
Key Points

1. Tree
2. Breadth-First and Depth-First Traversal
3. Binary Tree

Tree

Basic Definition — Children

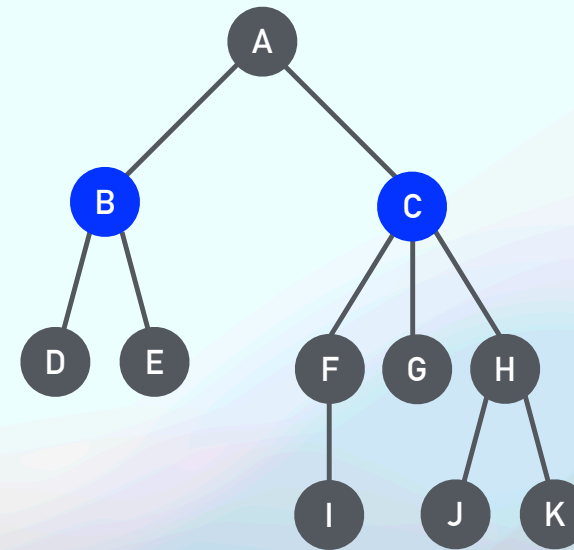
A	
C	
K	



Tree

Basic Definition — Children

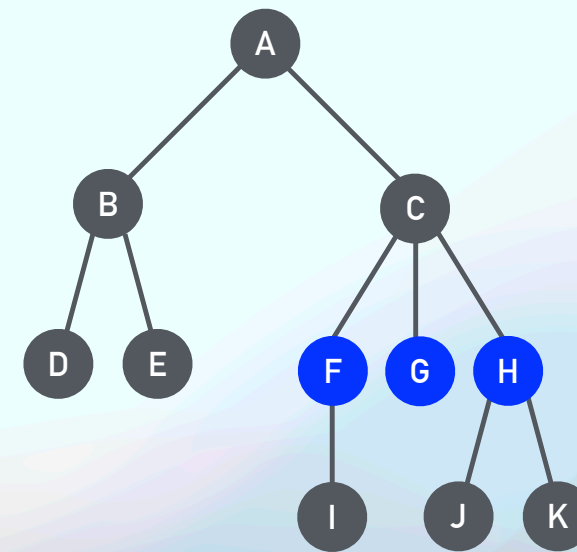
A	BC
C	
K	



Tree

Basic Definition — Children

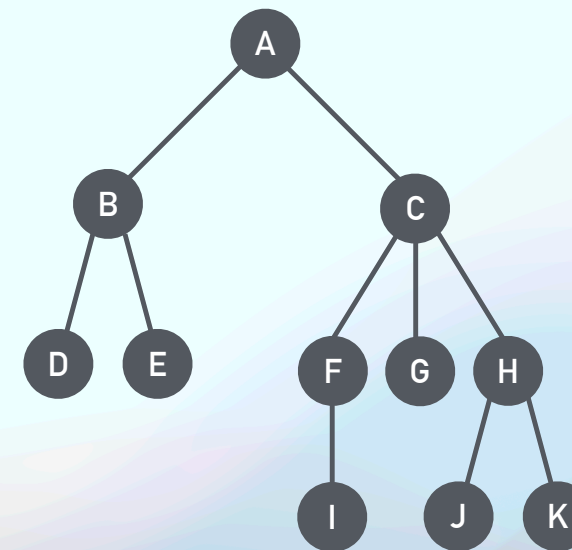
A	BC
C	FGH
K	



Tree

Basic Definition — Children

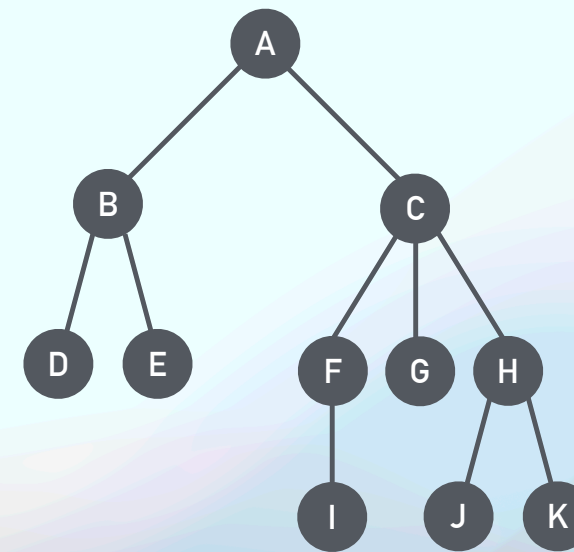
A	B C
C	F G H
K	None



Tree

Basic Definition — Parent

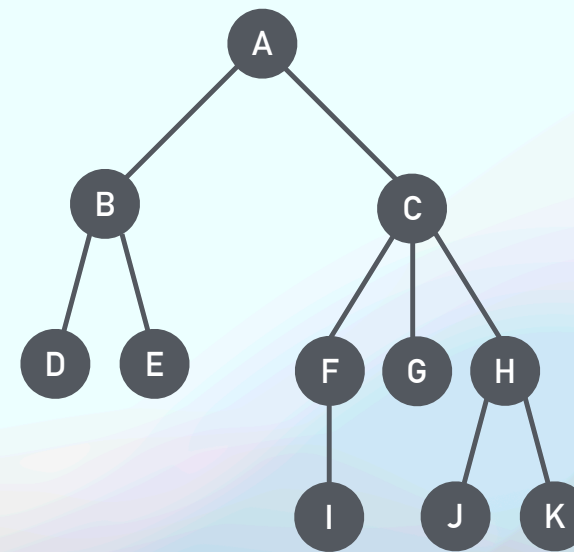
A	
C	
K	



Tree

Basic Definition — Parent

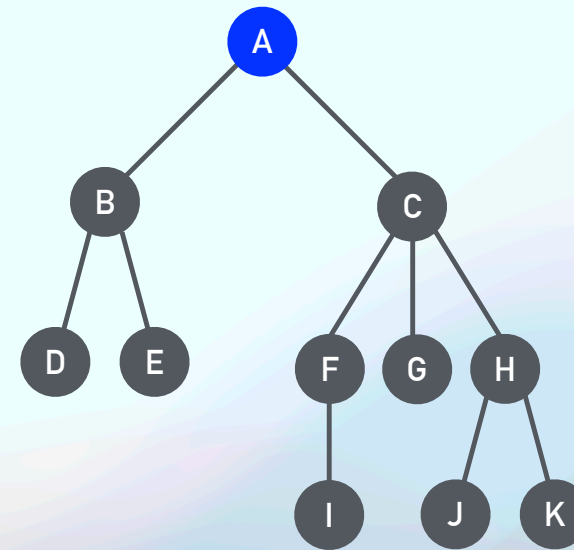
A	None
C	
K	



Tree

Basic Definition — Parent

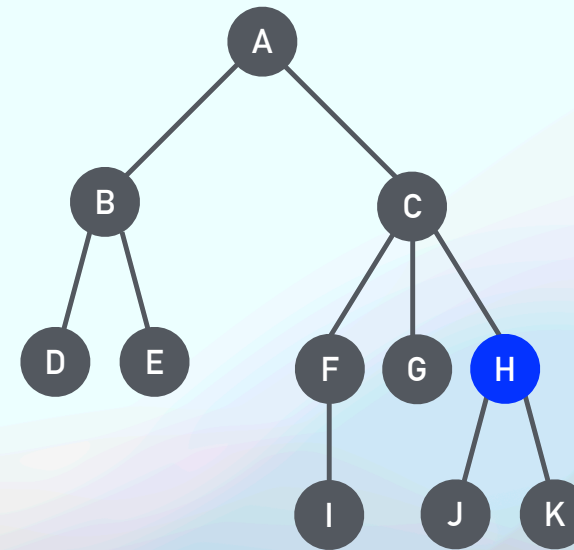
A	None
C	A
K	



Tree

Basic Definition — Parent

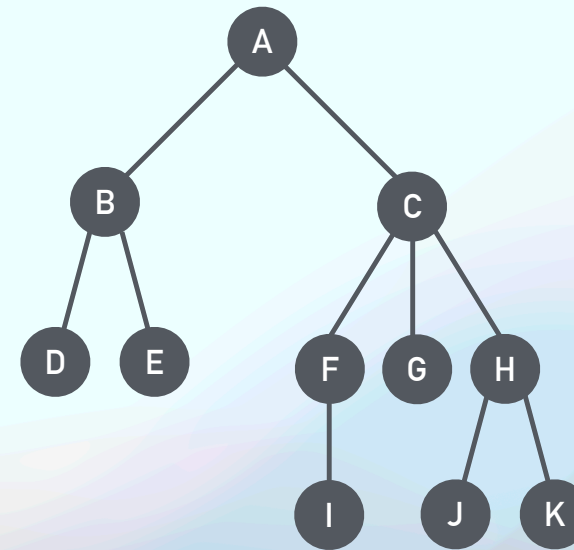
A	None
C	A
K	H



Tree

Basic Definition — Degree

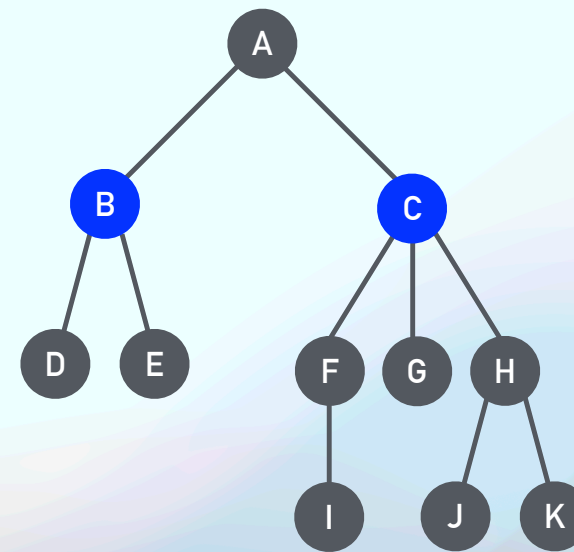
A	
C	
K	



Tree

Basic Definition — Degree

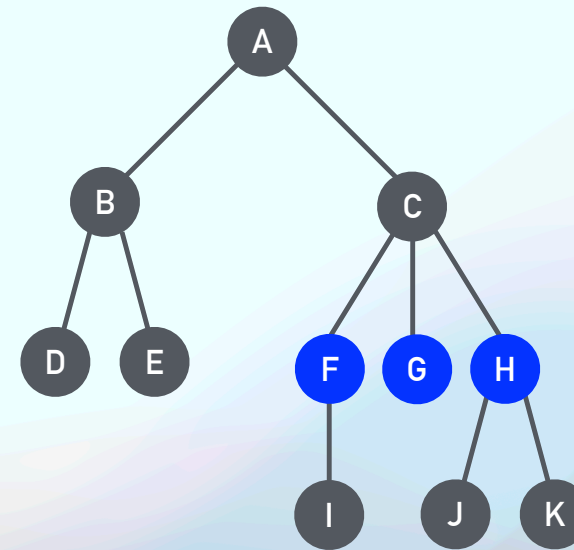
A	2
C	
K	



Tree

Basic Definition — Degree

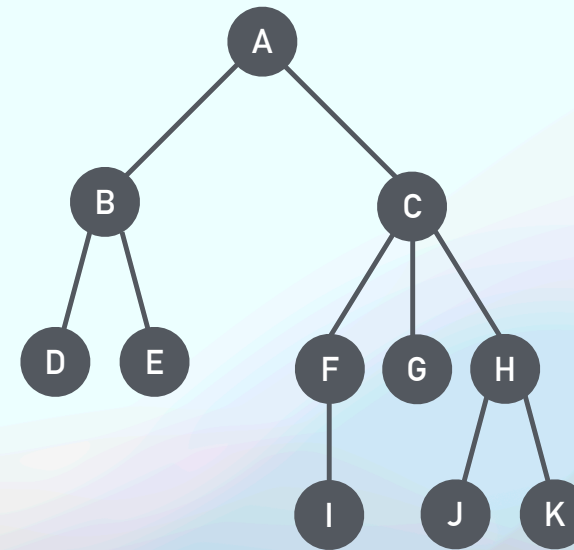
A	2
C	3
K	



Tree

Basic Definition — Degree

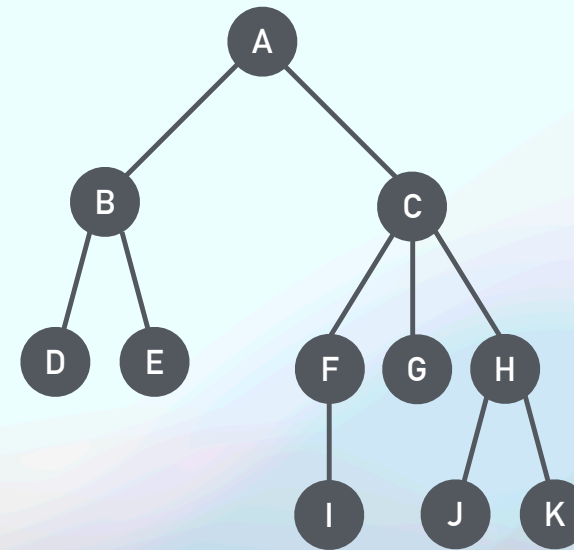
A	2
C	3
K	0



Tree

Basic Definition — Siblings

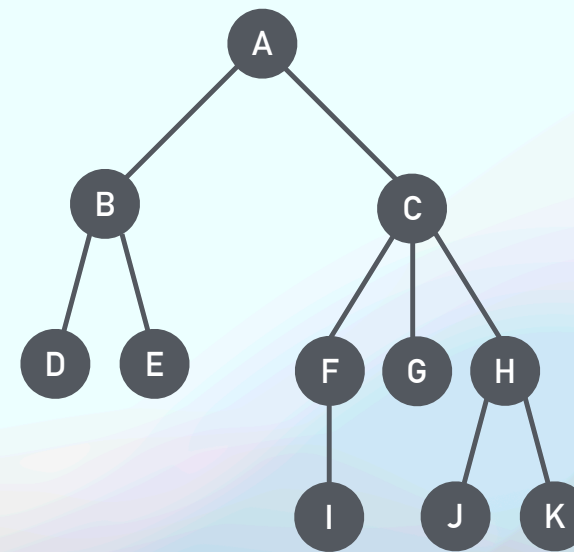
A	
C	
K	



Tree

Basic Definition — Siblings

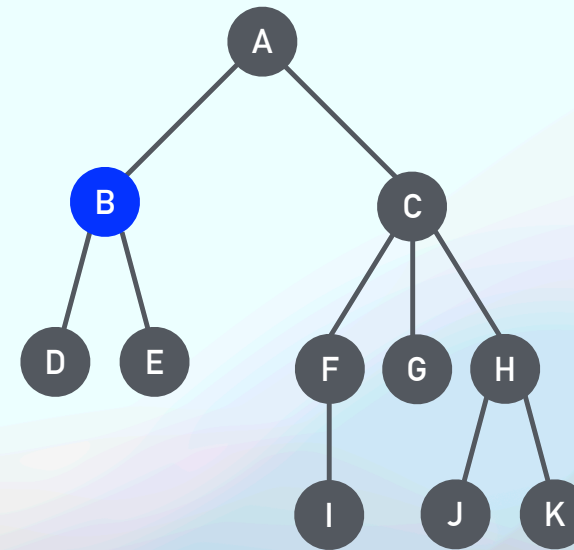
A	None
C	
K	



Tree

Basic Definition — Siblings

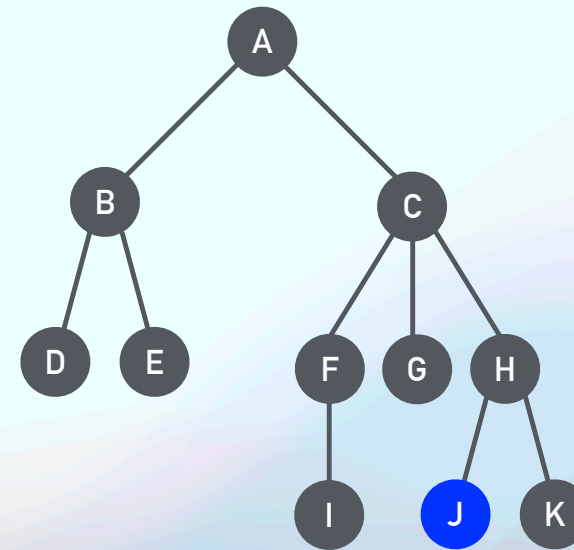
A	None
C	B
K	



Tree

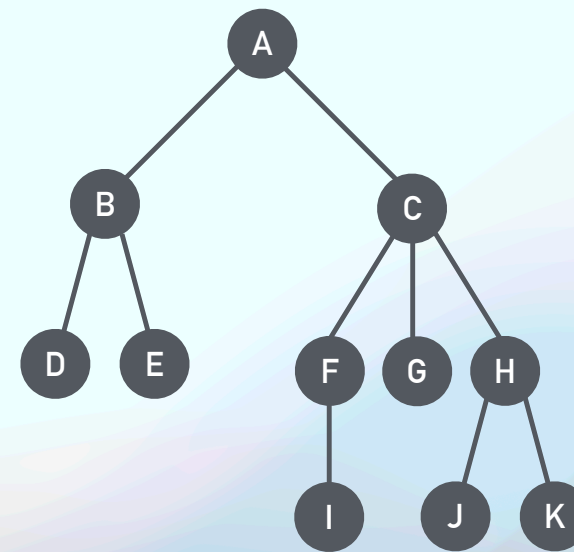
Basic Definition — Siblings

A	None
C	B
K	J



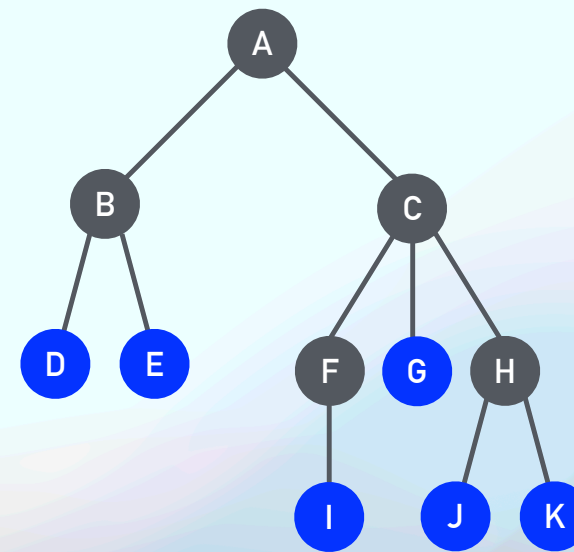
Tree

Basic Definition — Leaf nodes



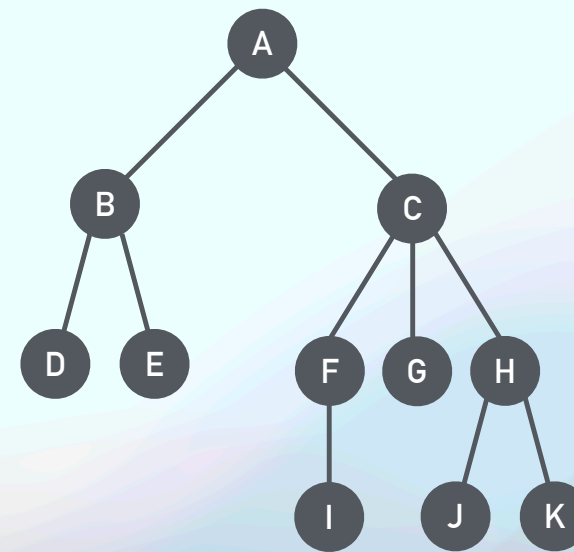
Tree

Basic Definition — Leaf nodes



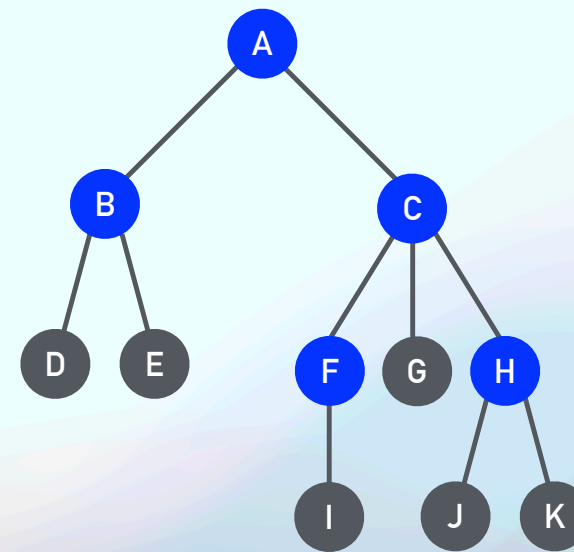
Tree

Basic Definition — Internal nodes



Tree

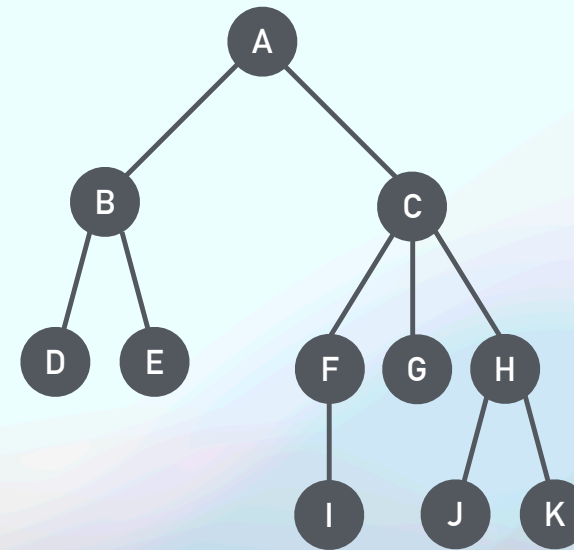
Basic Definition — Internal nodes



Tree

Basic Definition — Path

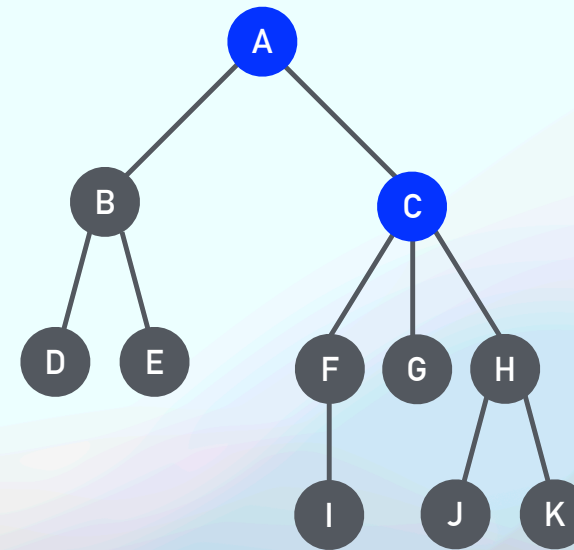
A-C	
A-J	
C-K	



Tree

Basic Definition — Path

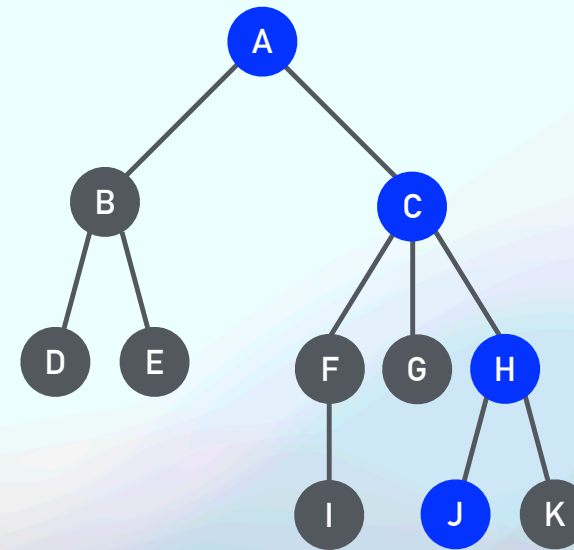
A-C	A-C
A-J	
C-K	



Tree

Basic Definition — Path

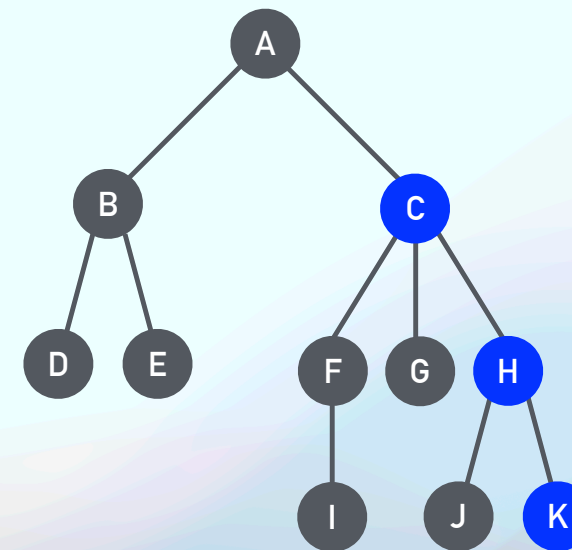
A-C	A-C
A-J	A-C-H-J
C-K	



Tree

Basic Definition — Path

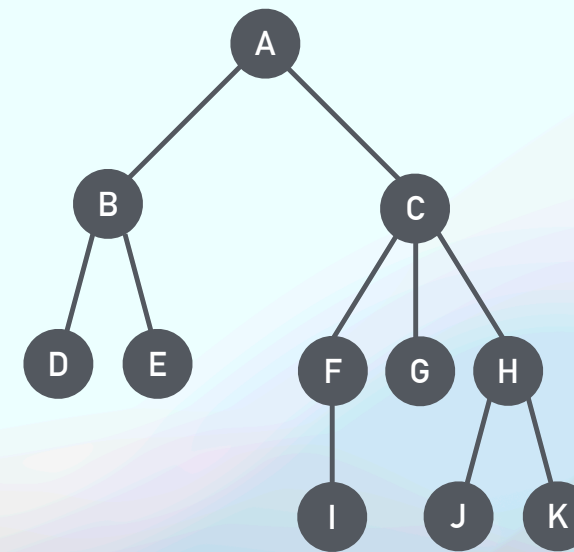
A-C	A-C
A-J	A-C-H-J
C-K	C-H-K



Tree

Basic Definition — Depth, Height

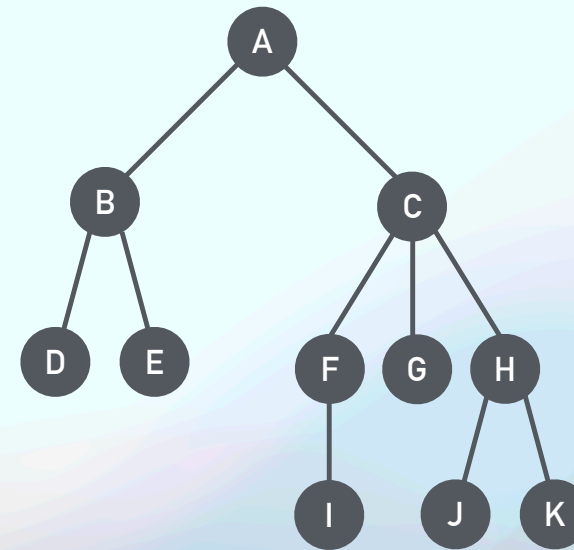
A	
C	
K	



Tree

Basic Definition — Depth, Height

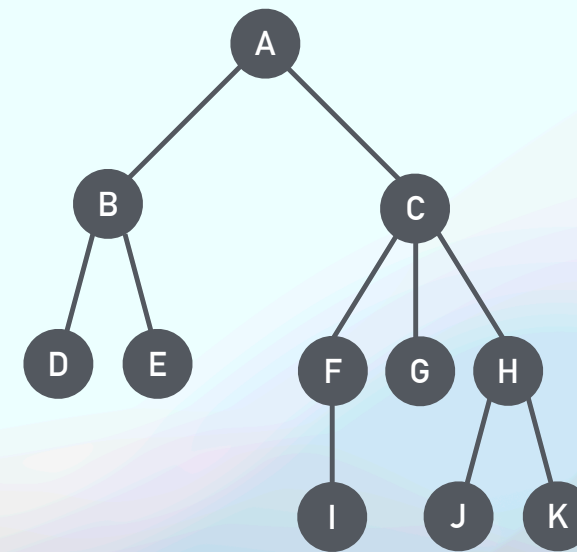
A	0
C	
K	



Tree

Basic Definition — Depth, Height

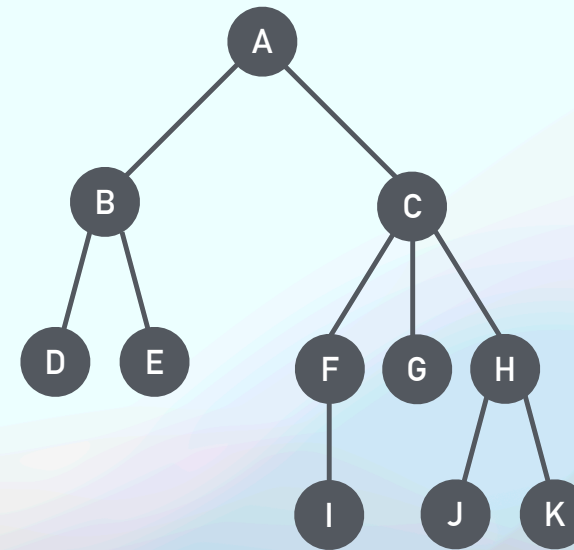
A	0
C	1
K	



Tree

Basic Definition — Depth, Height

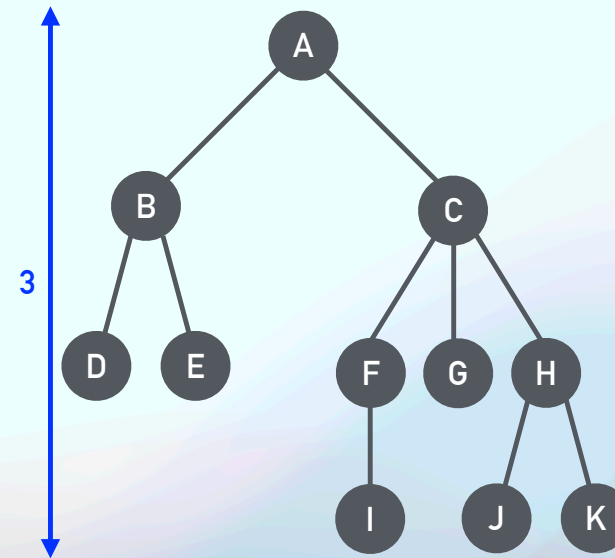
A	0
C	1
K	3



Tree

Basic Definition — Depth, Height

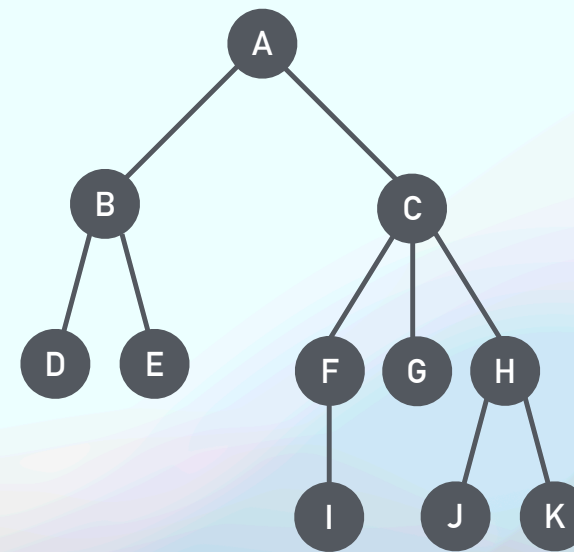
A	0
C	1
K	3



Tree

Basic Definition — Ancestor

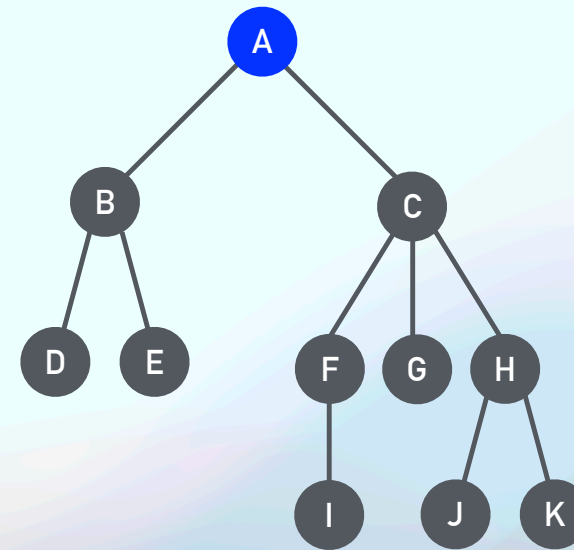
A	
C	
K	



Tree

Basic Definition — Ancestor

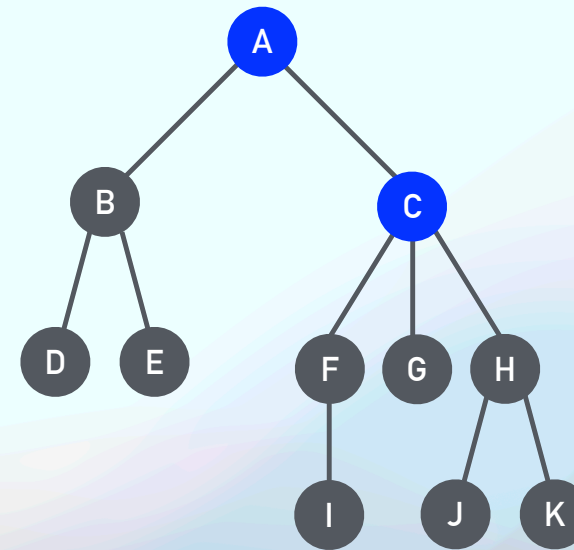
A	A
C	
K	



Tree

Basic Definition — Ancestor

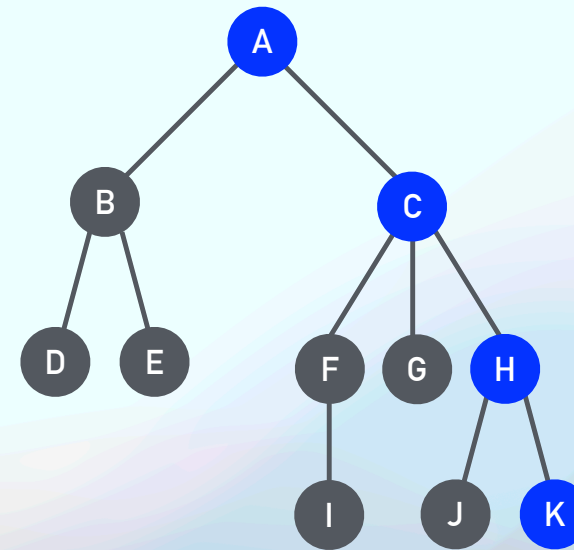
A	A
C	A C
K	



Tree

Basic Definition — Ancestor

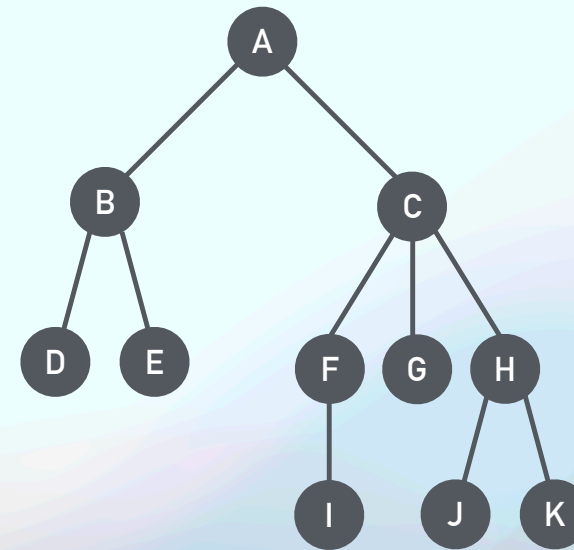
A	A
C	A C
K	A C H K



Tree

Basic Definition — Descendant

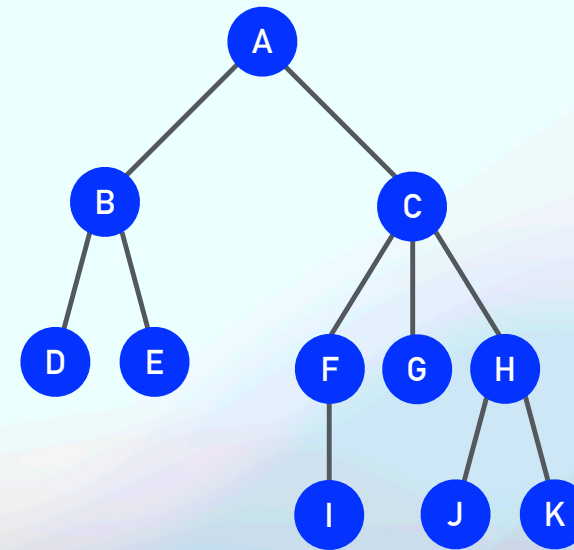
A	
C	
K	



Tree

Basic Definition — Descendant

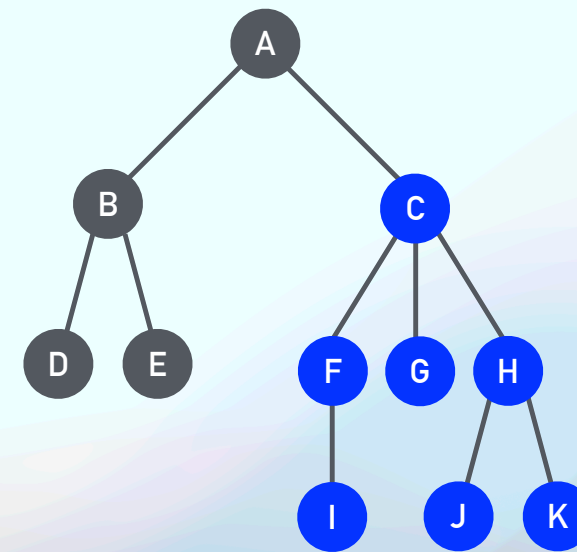
A	ABCDEFGHIJK
C	
K	



Tree

Basic Definition — Descendant

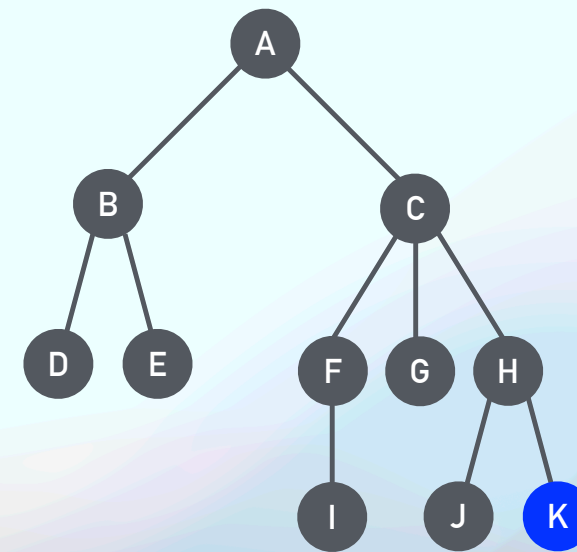
A	ABCDEFGHIJK
C	CFGHIJK
K	



Tree

Basic Definition — Descendant

A	ABCDEFGHIJK
C	CFGHIJK
K	K



CS101-Quiz5-Review

Key Points

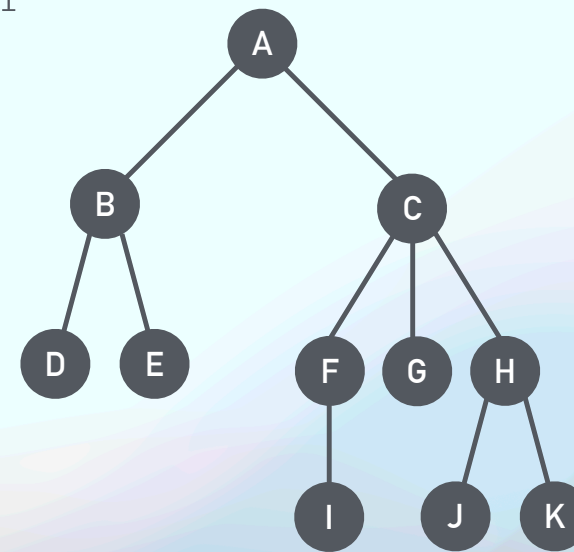
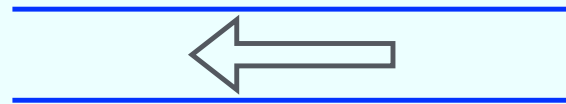
1. Tree

2. Breadth-First and Depth-First Traversal

3. Binary Tree

Breadth-First Traversal

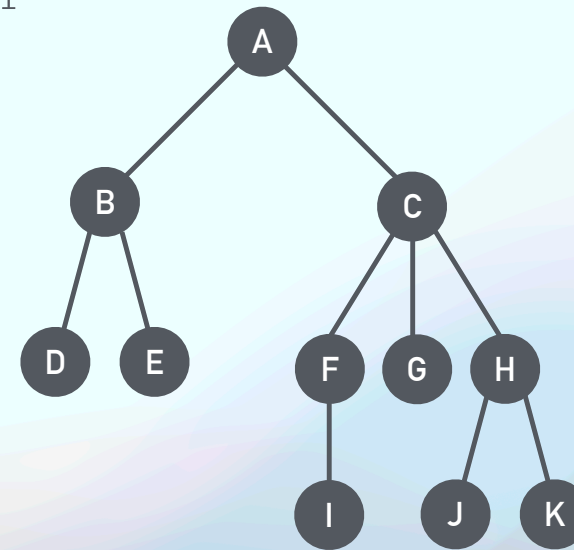
Queue



Breadth-First Traversal

Queue

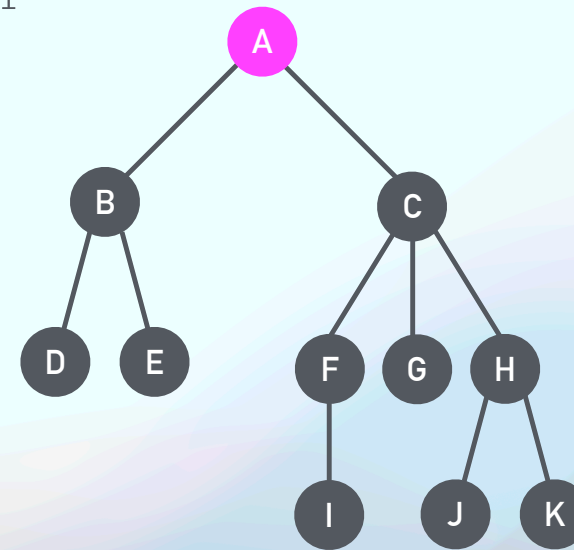
A



Breadth-First Traversal

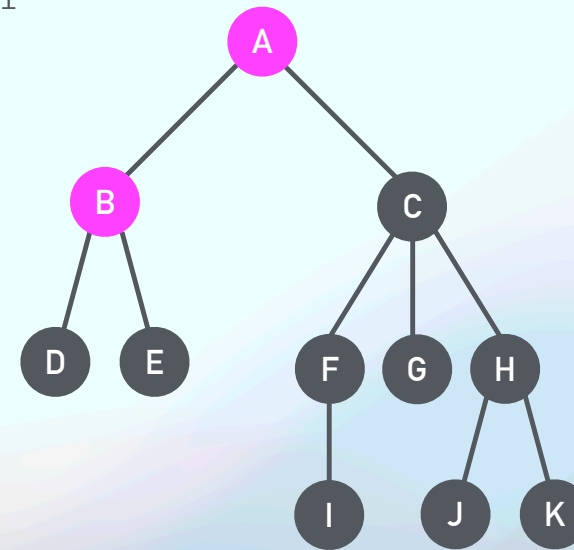
Queue

B C



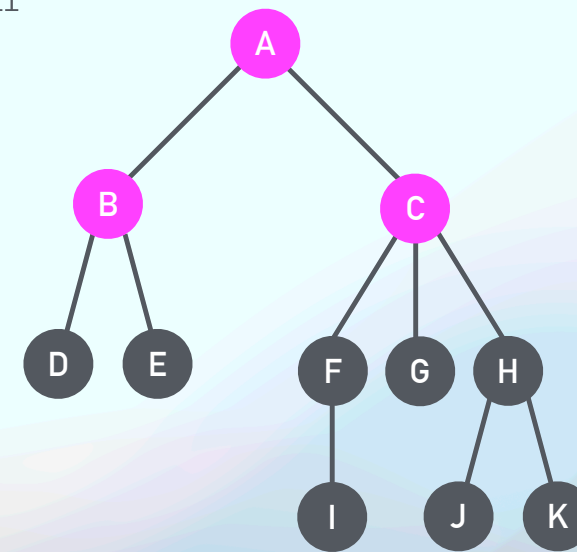
Breadth-First Traversal

Queue



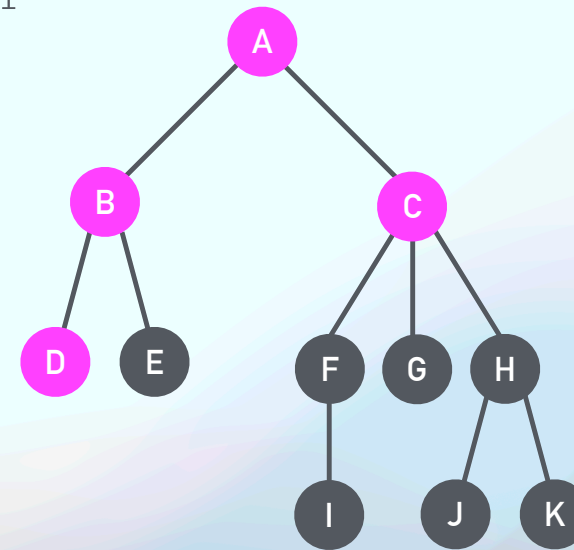
Breadth-First Traversal

Queue



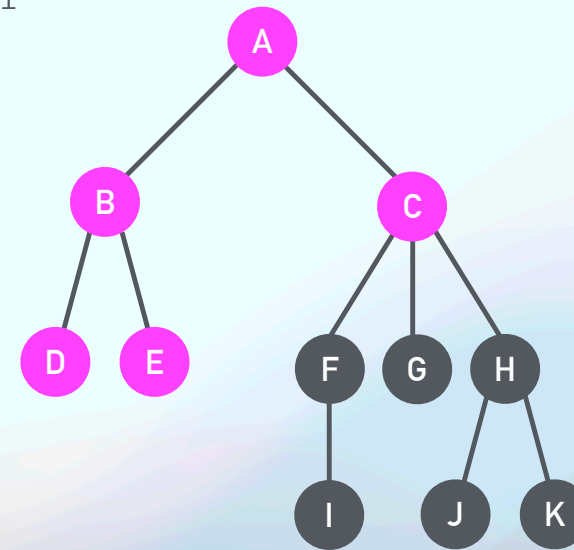
Breadth-First Traversal

Queue



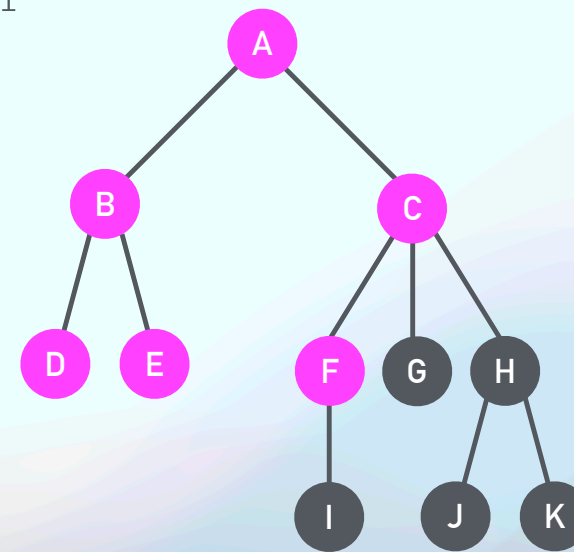
Breadth-First Traversal

Queue



Breadth-First Traversal

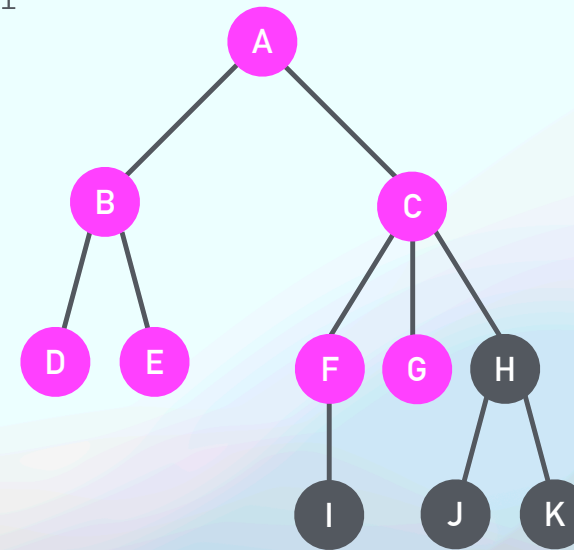
Queue



Breadth-First Traversal

Queue

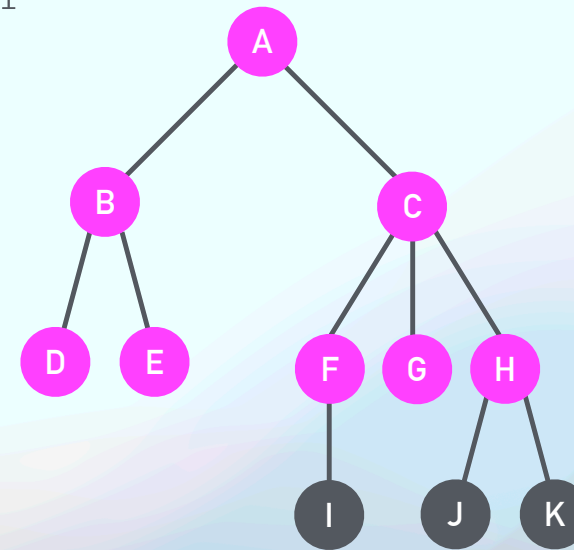
H I



Breadth-First Traversal

Queue

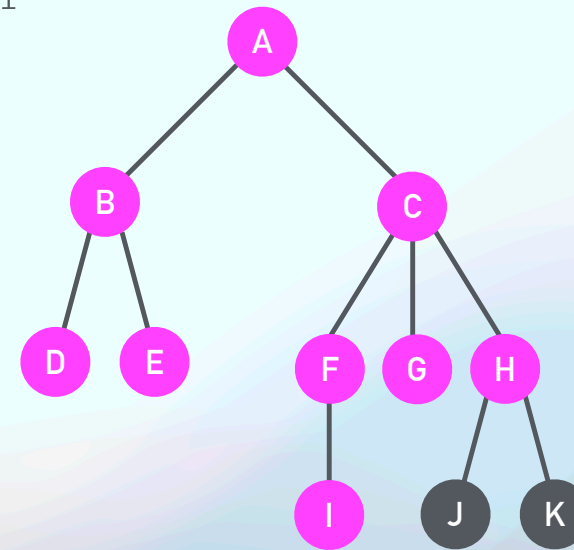
I J K



Breadth-First Traversal

Queue

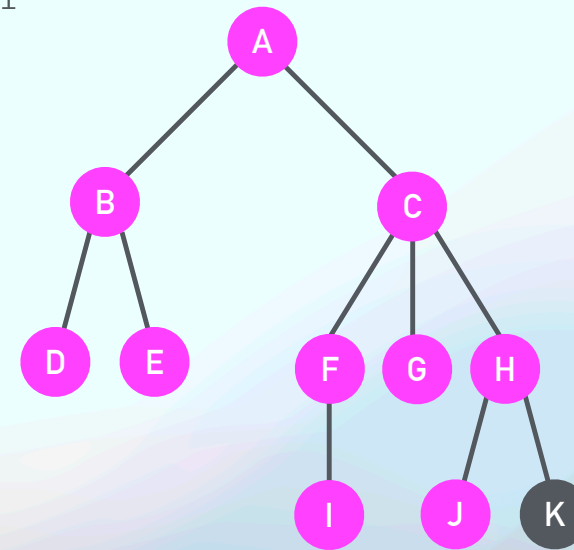
J K



Breadth-First Traversal

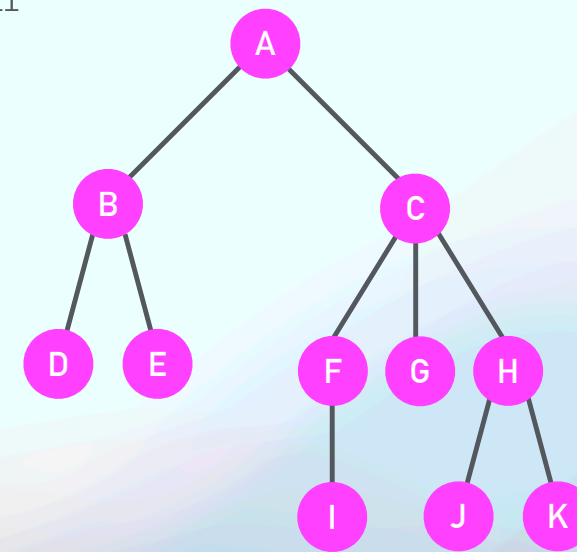
Queue

K



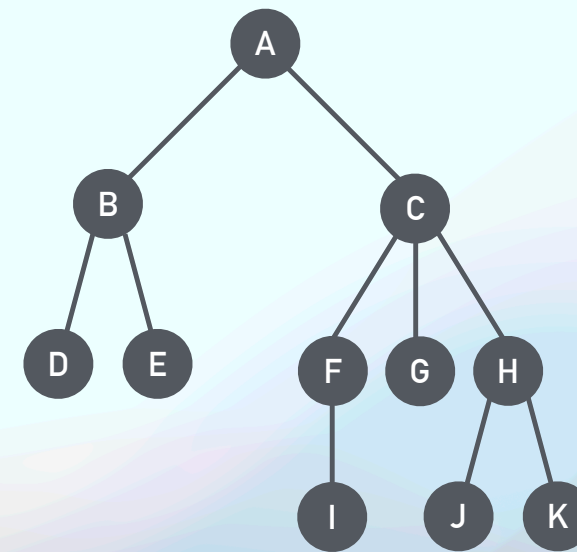
Breadth-First Traversal

Queue



Depth-First Traversal

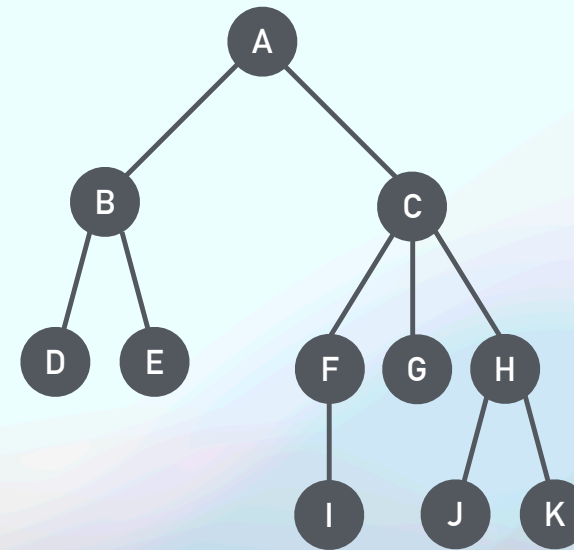
Stack



Push with alphabetical order

Depth-First Traversal

Stack



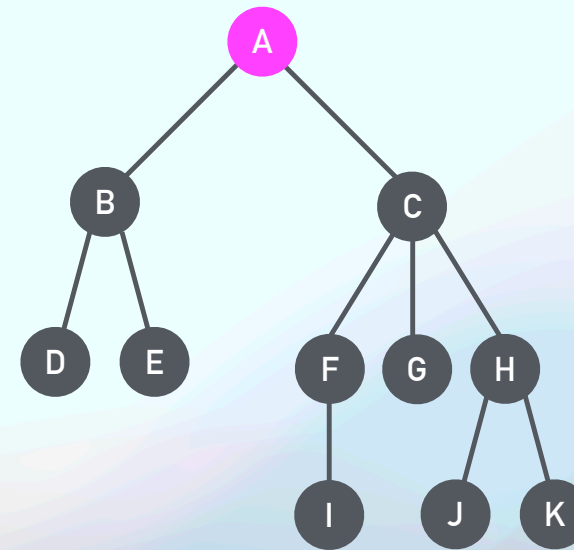
Push with alphabetical order

Depth-First Traversal

Stack



Push with alphabetical order

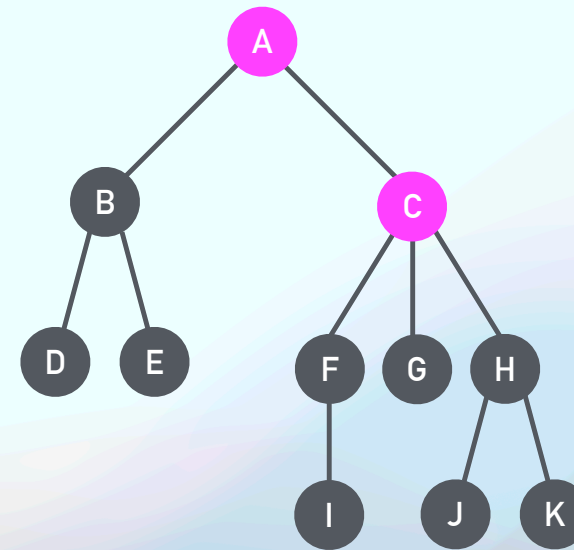


Depth-First Traversal

Stack



Push with alphabetical order

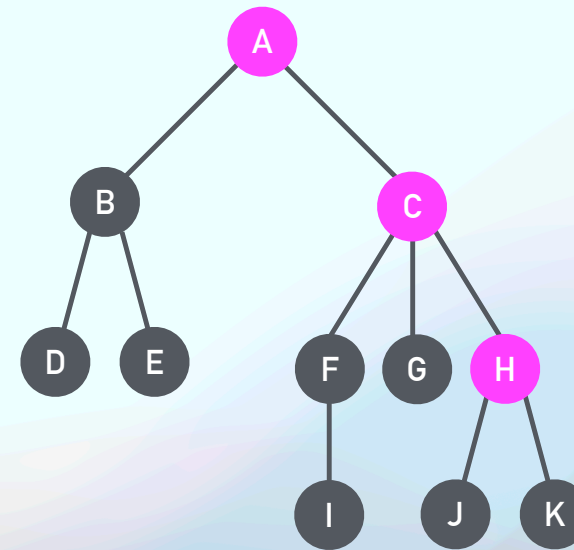


Depth-First Traversal

Stack



Push with alphabetical order

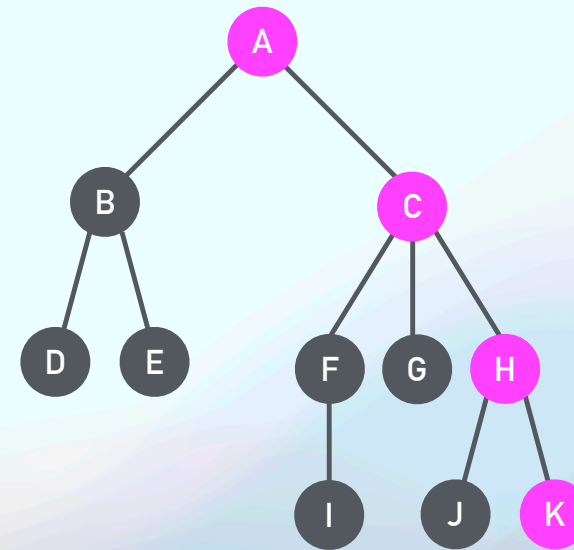


Depth-First Traversal

Stack



Push with alphabetical order

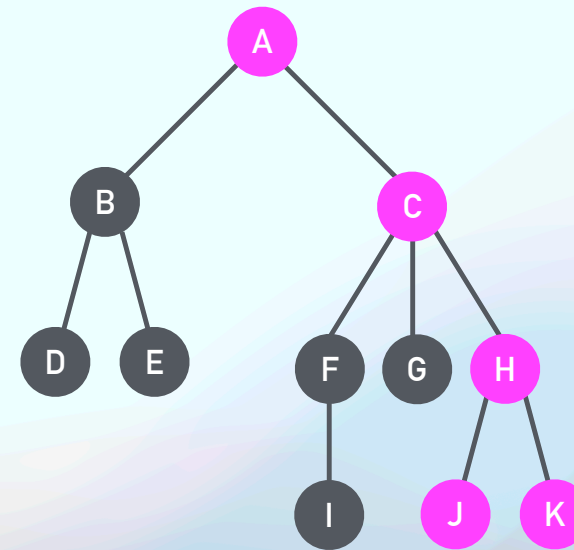


Depth-First Traversal

Stack



Push with alphabetical order

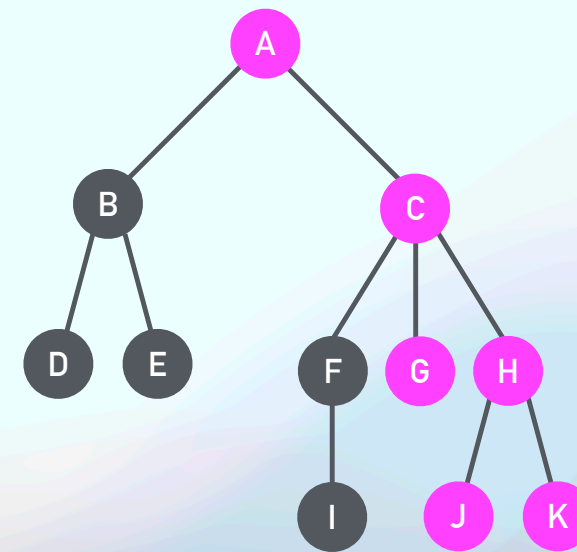


Depth-First Traversal

Stack

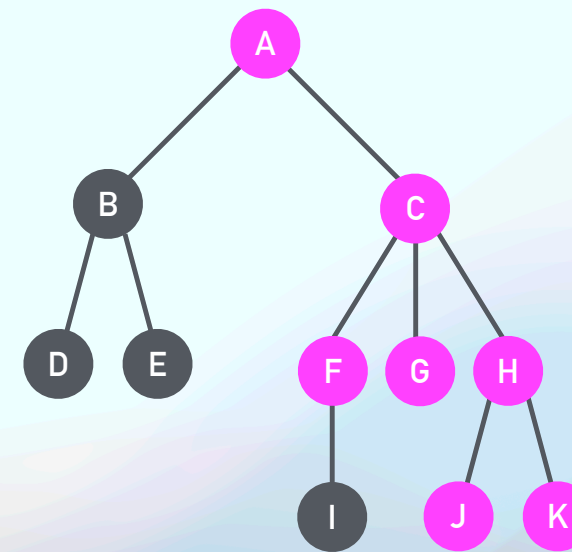


Push with alphabetical order



Depth-First Traversal

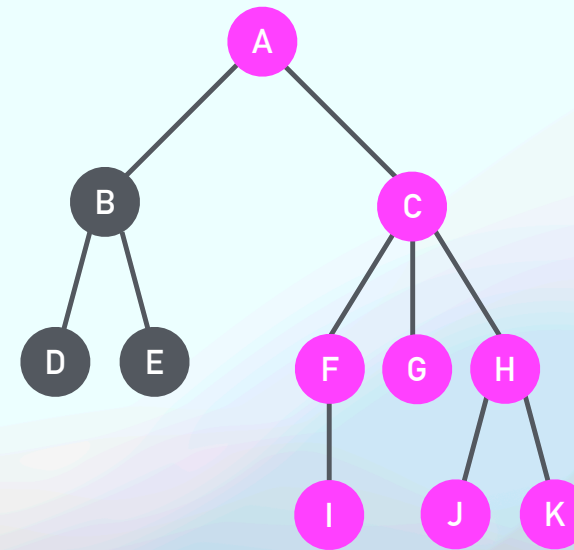
Stack



Push with alphabetical order

Depth-First Traversal

Stack



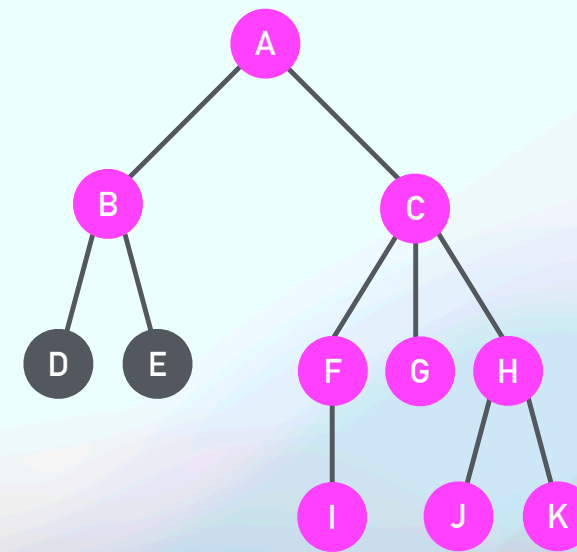
Push with alphabetical order

Depth-First Traversal

Stack

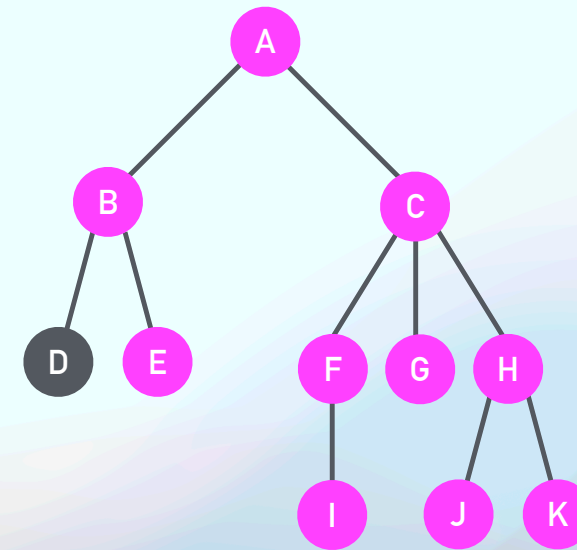


Push with alphabetical order



Depth-First Traversal

Stack



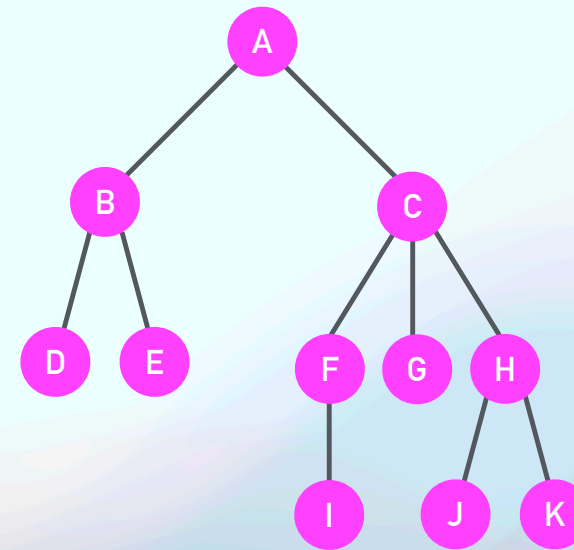
Push with alphabetical order

Depth-First Traversal

Stack



Push with alphabetical order



CS101-Quiz5-Review

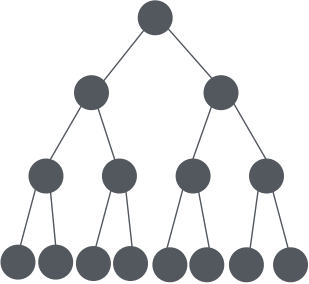
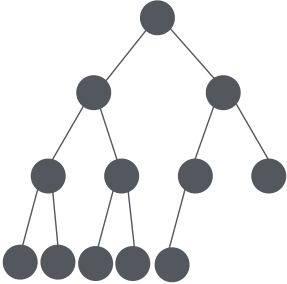
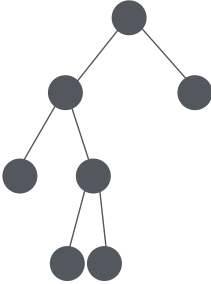
Key Points

1. Tree
2. Breadth-First and Depth-First Traversal

3. Binary Tree

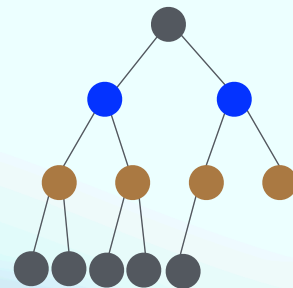
Binary Tree

Examples

Perfect Binary Tree	Complete Binary Tree	Full Binary Tree
		
"Perfect"	Breadth-First	Leaf or Full

Binary Tree

Complete Binary Tree — Array Storage



Binari
Comple

```
1 struct Node {
2     DataType m_data;
3     Node *parent;
4     Node *left;
5     Node *right;
6 };
7
8 DataType get_data(Node *n) {
9     return n->m_data;
10 }
11
12 Node *get_parent(Node *n) {
13     return n->parent;
14 }
15
16 Node *get_left(Node *n) {
17     return n->left;
18 }
19
20 Node *get_right(Node *n) {
21     return n->right;
22 }
```

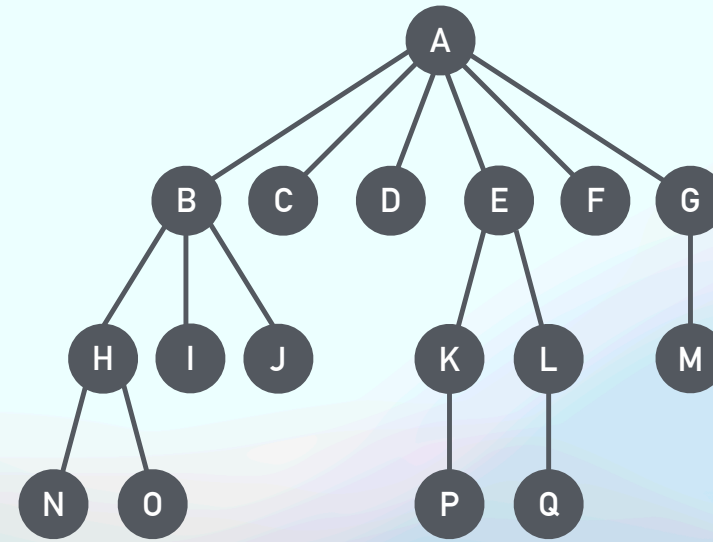
Store

```
1 DataType arr[SIZE];
2
3 size_t head = 0;
4
5
6
7
8 DataType get_data(size_t idx) {
9     return arr[idx];
10 }
11
12 size_t get_parent(size_t idx) {
13     return (idx - 1) / 2;
14 }
15
16 size_t get_left(size_t idx) {
17     return 2 * idx + 1;
18 }
19
20 size_t get_right(size_t idx) {
21     return 2 * idx + 2;
22 }
```

左边的会浪费 $3N * \text{sizeof(ptr)}$ 的空间，右边使用的空间取决于SIZE
因为cache的原因，右边的性能要更好。

Binary Tree

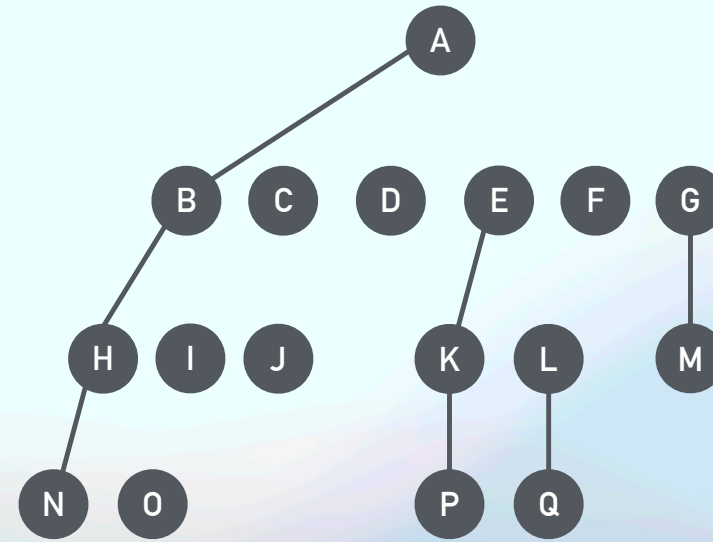
Left-child right-sibling



Knuth transform

Binary Tree

Left-child right-sibling



Binary Tree

Left-child right-sibling

